

Chapter 1

① Distance betⁿ two points $P_1(x_1, y_1)$ & $P_2(x_2, y_2)$

$$P_1P_2 = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

From origin, $P_1P_2 = \sqrt{x^2 + y^2}$

② Section, Internally, $x = \frac{m_1x_2 + m_2x_1}{m_1 + m_2}$, $y = \frac{m_1y_2 + m_2y_1}{m_1 + m_2}$

Externally, $x = \frac{m_1x_2 - m_2x_1}{m_1 - m_2}$, $y = \frac{m_1y_2 - m_2y_1}{m_1 - m_2}$

middle point, $x = \frac{x_1 + x_2}{2}$, $y = \frac{y_1 + y_2}{2}$

③ Area, $\Delta = \frac{1}{2} \begin{vmatrix} x_1 & y_1 \\ x_2 & y_2 \\ \vdots & \vdots \\ x_n & y_n \\ x_1 & y_1 \end{vmatrix}$

$$R(x, y) = R(p, -f)$$

points (a, b) , (a', b') , (a'', b'') are collinear, show that $d_1a = d_2a' = d_3a''$

$$0 = \begin{vmatrix} a & a' & a'' \\ b & b' & b'' \\ 1 & 1 & 1 \end{vmatrix} = 0$$

$$0 = \{ (a'b'' - a''b') + (a''a' - a'a'') + (a'a - a'a) \}$$

$$a - a' + a'' = 0$$

* For what value of k , are the points $(2, 3)$, $(-4, -6)$ & $(k, 12)$ Colinear.

Ans: $\therefore \frac{1}{2} \begin{vmatrix} 2 & -4 & k \\ 3 & -6 & 12 \end{vmatrix} = 0$

$$D\{(-12+12) + (-48+6k) + (3k-24)\} = 0$$

$$\Rightarrow 9k - 72 = 0$$

$$\boxed{k = 8}$$

* Co-ordinates of point $R(x, y)$ which divides the joint of points $P(x_1, y_1)$ and $Q(x_2, y_2)$ externally in ratio $\frac{m_1}{m_2} = \frac{8}{3}$

Ans: $x = \frac{8 \cdot 3 - 3(-2)}{8-3}$, $y = \frac{8(-4) - 3(1)}{8-3}$

$$x = \frac{30}{5}, y = \frac{-35}{5}$$

$$x = 6, y = -7$$

$$\boxed{R(x, y) = R(6, -7)}$$

* If the points (a, b) , (a', b') , $(a-a', b-b')$ are colinear, show their joint passes through the origin and that $ab' = a'b$

Ans: $\frac{1}{2} \begin{vmatrix} a & a' & a-a' \\ b & b' & b-b' \end{vmatrix} = 0$

$$\Rightarrow \{ (ab' - a'b) + (a'b - a'b' - ab' + a'b') + (ab - a'b = ab + ab') \} = 0$$

$$\Rightarrow -a'b + ab' = 0$$

$$\boxed{ab' = a'b}$$

* The coordinates of A, B, C, D are A(6, 3), B(-3, 5), C(4, -2), D(x, 3x) respectively & $\frac{\Delta DBC}{\Delta ABC} = \frac{1}{2}$, when Δ is the area of Δ . Find x.

Ans: $\Delta DBC = \frac{1}{2} \begin{vmatrix} x & -3 & 4 \\ 3x & 5 & -2 \end{vmatrix}$

$$= \frac{1}{2} \{ (5x + 9) + (6 - 20) + (12x + 2) \}$$

$$= \frac{1}{2} (28x - 14)$$

$$\Delta ABC = \frac{1}{2} \begin{vmatrix} 6 & -3 & 4 \\ 3 & 5 & -2 \end{vmatrix}$$

$$= \frac{1}{2} \{ (+30 + 9) + (6 - 20) + (12 + 12) \}$$

$$= \frac{1}{2} (39 - 14 + 24)$$

$$= \frac{1}{2} (49)$$

$$\frac{\Delta DBC}{\Delta ABC} = \frac{1}{2}$$

$$\Rightarrow \frac{\frac{1}{2} (28x - 14)}{\frac{1}{2} (49)} = \frac{1}{2}$$

$$\Rightarrow 56x - 28 = 49$$

$$\Rightarrow 56x = 77$$

$$\boxed{x = 11/8}$$

